

Agenda.

→ Features / MVP ⇒ Functional Req.

→ Scale Estimation |

back of the envelope calculation.

→ Design Tradeoffs. ⇒ Non Functional Req.

→ Design Deepdive.

Evaluate HashMap.

↓
key value DB like REDIS.

↓
Automatically sharded based key.

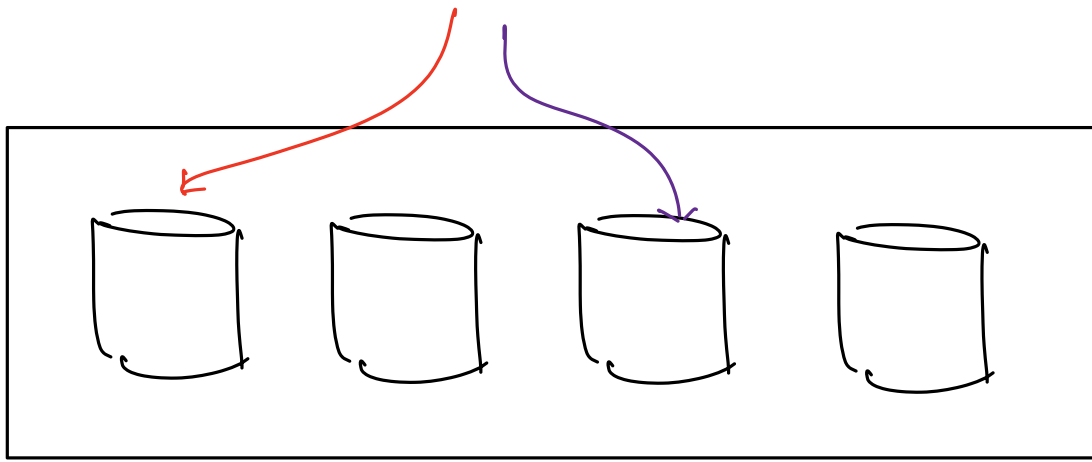
frequency HM.

"microsoft" : 100K + 1

"scalex" : 20K + 1

"google" : 50K

—
—
—
—



Suggestions HM

prefix : Top 5 suggestions for
the given prefix.

"mic" :

microservice : 100K
microsoft : 99K
microname : 40K
micromax : 10K

 +++++
10001

String : "microsoft:100K; microservice:50K;..."

of Write = 15 B / Day.

of Reads = 75 B / Day.

Write

"microsoft"

freq HM.

Increment
the freq.

IntraShard
query.

microsoft : 100k+1

mic →

micro → microsoft : 100k+1

micro -

micros -

microso -

microsoft -

⇓

InterShard X

Read

IntraShard.

getSuggestions("mic")

suggestionHM("mic")



"mic" :

microservice : 100k
microsoft : 99k → +++++
microname : 40k
micromax : 10k

Write $\Rightarrow$ Inter Shard

Read $\Rightarrow$ Intra Shard.

$\Rightarrow$ Both R+W heavy.

$\Rightarrow$ Batch Processing $\rightarrow$ Async



Combine write operations & perform them in one go.

HM-1
↳ Freq HM

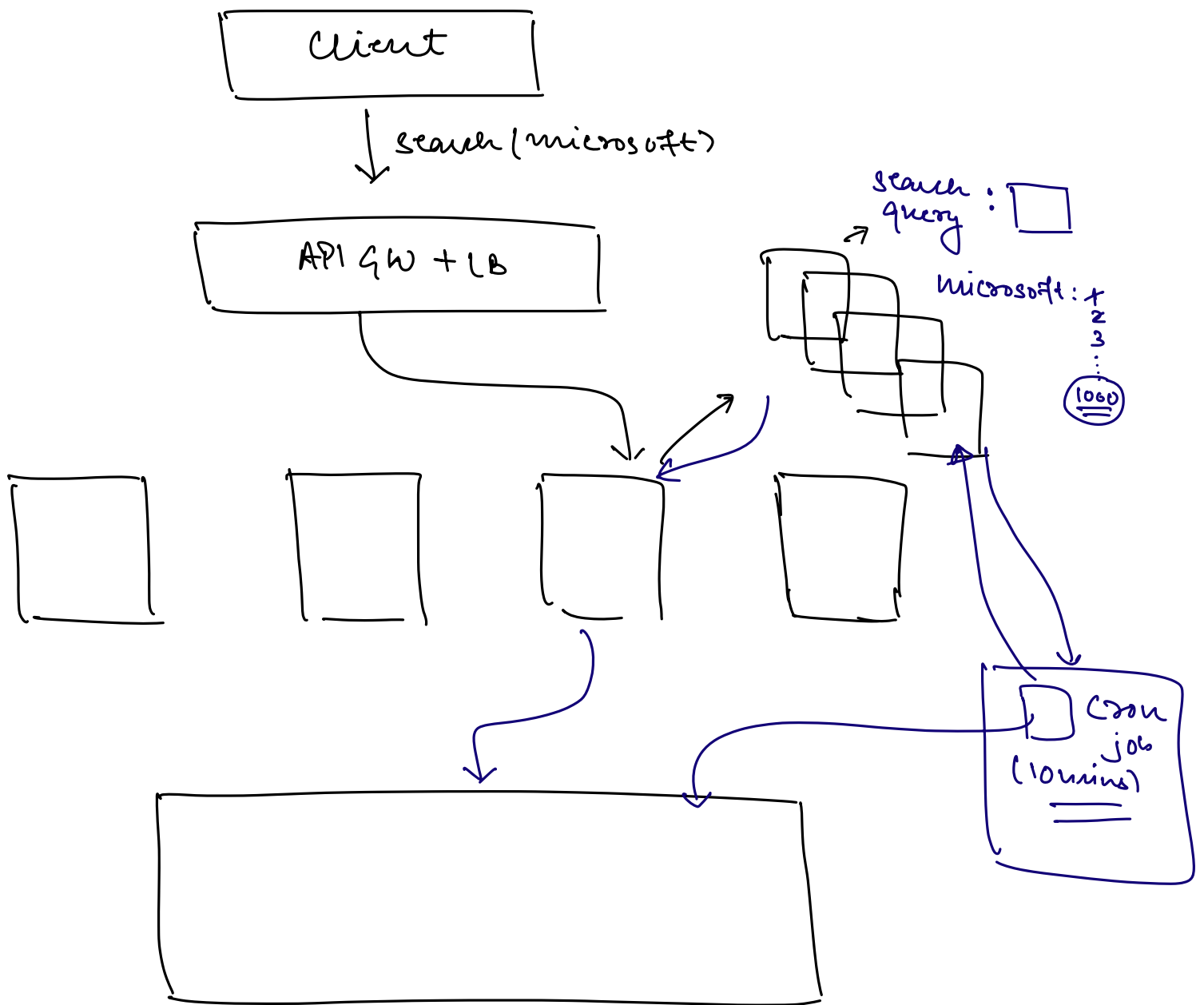
HM-2
↳ Suggestions HM
(Preprocessed Data)

$\Rightarrow$ Batch Processing



Async Updates

$\Rightarrow$ Eventual Consistency.



15 B writes / Day.

$$\downarrow$$

$$\frac{15 \times 10^6}{1000} = \underline{\underline{15M}}$$

75 B reads / Day.

R+W heavy $\Rightarrow$ Read Heavy.

$\Rightarrow$ Cache the top suggestions globally.

Sampling.

$\downarrow$
Pols.